

Comparative genomics reveals the evolution of antimicrobial resistance in *Bacteroides nordii*

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ABSTRACT

Bacteroides nordii, is an understudied member of the pathogenic *B. fragilis* group which comprises several multidrug-resistant (MDR) strains. Thus, it is of great interest to study the genome biology of *Bacteroides nordii*. However, no detailed study is available that characterized *B. nordii* at the genetic level and explored its role as a potential pathogen. We isolated an MDR strain viz., *B. nordii* PGMM4098 from the pus sample and subjected it to whole genome sequencing using Illumina technology. The draft genome was *de-novo* assembled and annotated, followed by comprehensive comparative genomics analyses using the publicly available genome dataset of *B. nordii*. The pan-genome analysis revealed the open nature of *B. nordii*, indicating the continuous accumulation of novel genes in non-core components leading to the emergence of new strains of this species. The thirteen antimicrobial resistance (AMR) genes identified in the genomes of all *B. nordii* strains were part of the non-core component of the pan-genome. Of these, four AMR genes, *nimE*, *aadS*, *mef(En2)*, and *ermB/F/G* were found to be acquired via the process of horizontal gene transfer (HGT) from anaerobic Bacteroidetes. Importantly, the *nimE* gene conferring metronidazole resistance was found to be present only in *B. nordii* PGMM4098, which harbors five other AMR genes encoded in its genome. Of these, *nimE* (metronidazole resistance), *ermB/F/G* (macrolide-lincosamide-streptogramin B resistance), and *cfxA2/A3* (class A β -lactam resistance) genes were further validated using targeted polymerase chain reaction assay. Notably, these three genes were also found to be under the operation of positive selective pressure suggesting the diversification of these genes, which might lead to the emergence of new MDR strains of *B. nordii* in the near future. Our study reported and characterized the genome of the first MDR strain of *B. nordii* and revealed the AMR evolution in this species using a comprehensive comparative genomics approach.

1. Introduction

The genus *Bacteroides* comprises Gram-negative bacilli that form a significant part of the human gut microbiota and possesses both probiotic and pathogenic potential. The members of the genus, predominantly the *B. fragilis* group (*B. fragilis*, *B. thetaiotaomicron*, etc.), have been associated with life-threatening conditions when displaced from their local habitat, as seen in patients with compromised mucosal integrity and lowered immunity [1,2]. *Bacteroides nordii* is not a member of the *B. fragilis* group; however, it is closely related to *Bacteroides thetaiotaomicron* with at least 94% 16S rRNA gene sequence identity [3]. *B. nordii*, named after Carl Erik Nord, is an obligate anaerobe that has a cell size of 0.8–1.5 μ m by 0.5–5.0 μ m and grows as circular, convex, opaque grey colored colonies of 1–2 mm diameter.

Multidrug resistance (MDR) in *B. fragilis* group is already known, and whole-genome sequencing (WGS) has been employed to identify genes/mechanisms of antimicrobial resistance (AMR) in this group [4]. However, MDR in *B. nordii* is not known to date. A few genome studies have recently been carried out on various pathogenic *Bacteroides* spp., viz. *B. thetaiotaomicron* and *B. fragilis* [4,5]. However, no study is available which has characterized *B. nordii* at the genetic level and explored its role as a potential pathogen. Being a member of a pathogenic genus, the exploration of the genome evolution of *B. nordii* is of great interest. At the time of our analysis, the genome sequences of only twelve strains of *B. nordii* were publicly available.

In recent times, the AMR burden has been increasing rapidly worldwide [6]. Unmonitored usage of antimicrobials has further complicated this problem, especially in developing countries. The use of

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antimicrobial agents is a primary driver of AMR evolution leading to emergence of new AMR phenotypes and strains resistant to most of the current treatment regimens. The mobilome-mediated transfer of genetic material has played an important role in developing the global public health crisis of AMR. For instance, the *ermF* gene, which confers clindamycin resistance in anaerobes, is usually located on transferable plasmids along with genes encoding resistance to tetracycline [7]. The *ermF* gene was initially described in *Bacteroides* spp.; however, the GC content of the *Bacteroides* chromosome is 42% which is different from the GC content of the encoded *ermF* gene (34%) indicative of horizontal gene transfer (HGT) of this gene [8]. The horizontally acquired genes evolve under the operation of certain selective pressure and thus can propagate to the new generation [9,10]. It is important to note that genes without any selective pressure may get eliminated from a bacterial genome in the succeeding generations. Evolutionary studies are important to analyze the emergence and propagation of AMR in pathogens. For example, a recent study on *Klebsiella pneumoniae* revealed the transferability and evolution of *bla_{NDM-1}* and *bla_{KPC-2}* in clinical settings [11]. However, the lack of genome studies on *B. nordii* limits our knowledge of the AMR evolution in this pathogen. This necessitates comparative studies focusing on the genome and AMR evolution in *B. nordii*.

The genome evolution of a microorganism is linked to its lifestyle in a given environment. It is a continuous process that leads to significant variation in the microbial genome. During this process, microbes undergo the acquisitive or reductive mode of genome evolution. Pan-genome is a promising method to study and characterize the genome evolution of pathogens [10,12]. Pan-genome consists of core and accessory genes. The core genes of the microorganisms represent their minimal encoding potential which is essential for their survival and growth in a given environment. The accessory genes of the microorganisms provide them with selective advantages to adapt to a given ecological niche. Based on the characterization of the gene pool, the pan-genome is classified as open and closed. The open pan-genome indicates unlimited gene repertoire and continuous evolution of the taxa via different evolutionary mechanisms, viz., gene acquisition strategies, mutations, and diversification. On the other hand, a closed pan-genome indicates a limited gene pool that is less likely to acquire any additional functionalities even on the appearance of a new strain. Taxa such as *Escherichia coli*, *Cutibacterium acnes*, oral *Prevotella*, and the genus *Bifidobacterium* are known to harbor open pan-genomes, whereas species like *Bacillus anthracis* and *Bifidobacterium longum* have closed pan-genomes [13,14]. Similarly, the pan-genome of genus *Bacteroides* and *Parabacteroides* have also been determined but with a limited number of genomes for each species [15]. However, no study has determined the pan-genome of *B. nordii* to date.

In the present study, we carried out the genome characterization of the first MDR strain of *B. nordii* (PGMM4098) isolated from a clinical sample. The taxonomic positioning of *B. nordii* PGMM4098 was determined using phylogenetic analysis. In order to explore the genome potential of *B. nordii* in terms of growth, survival, and pathogenicity, we performed comprehensive comparative genomic analyses of this species. We further performed a pan-genome analysis of *B. nordii* to explore its evolutionary characteristics. The HGT and selective pressure analyses were also carried out for the exploration of AMR evolution in *B. nordii*.

2. Materials and methods

2.1. Sample collection and identification of MDR *B. nordii* isolate

The anaerobic bacteria were recovered from clinical samples submitted to the Clinical Bacteriology Laboratory of the Department of Medical Microbiology, PGIMER, Chandigarh, a tertiary care hospital that caters to nearly all North Indian states. The study includes one MDR *B. nordii* isolate that was identified out of 150 anaerobic bacteria tested against six antibiotics described in our previous study [2,16]. The MDR

B. nordii isolate was recovered from a brain abscess pus sample of a 14 years old male patient who had presented with left ear discharge for 18 months, diagnosed with a left temporoparietal space-occupying lesion. The brain abscess sample was processed for microbiological culture with the subsequent empirical antimicrobial treatment comprising of ceftriaxone, amikacin, and metronidazole.

The identification process included Gram stain microscopy and culture aerobically and anaerobically. Under anaerobic processing, the samples were inoculated in Robertson's cooked meat (RCM) broth as well as cultured on freshly prepared brucella agar plates supplemented with 5% laked sheep blood, hemin (5 mg/L), and vitamin K (1 mg/L). Metronidazole disc (5 µg) was placed at the center of each plate for presumptive testing of anaerobes and bore no therapeutic significance. Plates were subcultured anaerobically using two automated anaerobic gas generation systems; Don Whitley Scientific's Jar Gassing System and Anoxomat Mart II, Mart Microbiology BV, Lichtenvoorde, Netherlands. Anaerobiosis was created using the gas mixture (80% N₂, 10% CO₂, 10% H₂) and validated by a positive and negative control viz., *B. fragilis* ATCC 25285 and *Pseudomonas aeruginosa* ATCC 27853, respectively. After 48 h of anaerobic incubation, the plates were examined for a zone of inhibition around the metronidazole disc. The presence of a zone was suggestive of anaerobic growth, whereas, in the absence of the zone, the RCM turbidity and the corresponding aerobic plates were examined to ascertain the metronidazole-resistant anaerobic growth. We identified the bacterial isolates using MALDI-TOF MS; Biotyper 2.0 database (Bruker Daltonik GmbH, Bremen, Germany). The standard Bruker interpretative criteria were followed for identification with a score ≥2.0 considered accurate on species level; ≥ 1.7 but <2.0, on genus level, and <1.7, non-reliable identification (NRI).

2.2. Phenotypic antimicrobial susceptibility testing

The antimicrobial susceptibility testing (AST) was carried out using the breakpoint agar dilution method as per the Clinical and Laboratory Standards Institute (CLSI) guidelines (currently M11-A8) and the previously described method [17]. Bacterial isolates were tested against metronidazole, clindamycin, imipenem, piperacillin-tazobactam, cefoxitin, and chloramphenicol. The lowest concentration of the drug preventing bacterial growth was taken as the MIC and interpreted as per the CLSI breakpoints. A drug dilution was prepared for each antibiotic that contained on-scale values for the positive control (*B. fragilis* ATCC 25285).

2.3. Identification of AMR genes using molecular method

The polymerase chain reaction (PCR) assay was used to detect the presence of the genes conferring resistance to metronidazole (*nimE*), imipenem (*cfiA*), β-lactams (*cepA*), cefoxitin (*cfxA*), clindamycin (*ermF*), chloramphenicol (*cat*) and mobile genetic elements like insertion sequence IS1186 and the insertion sequence elements immediately upstream of the *cfiA* gene (*cfiA^{IS}*) that upregulates the expression of the *cfiA* gene. (Table S1).

2.4. DNA extraction

The DNA extraction was done using the QIAamp DNA Kits (Qiagen) as per the manufacturer's protocol for Gram-negative bacteria. The DNA concentration and purity were determined using the Qubit fluorometer (Thermo Fisher Scientific) and Nanodrop spectrophotometer (Thermo Fisher Scientific), respectively. The strain was stored at -80 °C in brain heart infusion broth with 40% (v/v) glycerol for future reference.

2.5. Whole genome sequencing

The genomic shotgun library was sequenced by the Clevergene Biocorp Pvt. Ltd., Bangalore, Karnataka, India using Illumina

sequencing technology. In brief, the WGS library preparation was carried out using the Nextera XT DNA Library Preparation Kit (Illumina, San Diego, USA) following the manufacturer's instructions, and the genome was sequenced on the Illumina HiSeq using paired-end chemistry.

2.6. Data preprocessing and genome assembly

After filtering low-quality reads ($<Q30$) and quality trimming using the PRINSEQ with default parameters, *de-novo* assembly of high-quality paired-end reads was conducted using the CLC Genomics Workbench v20.0 using standard settings.

2.7. Genome analysis

Rapid Annotation using Subsystem Technology (RAST) server [18] was used for annotation of the draft genome. Clusters of orthologous groups (COG) of proteins were inferred using a previously described method [19]. Results were parsed using an E-value cut-off of 10^{-5} for the best hit in the alignment. Pathway mapping was done using the BlastKOALA module of the Kyoto Encyclopedia of Genes and Genomes (KEGG) online server [20] using prokaryotes as the taxonomy group and genus prokaryotes as the KEGG genes database. Transporter proteins were predicted using Transporter Classification Database (TCDB) [21] by performing BLASTP (version 2.2.29+) with an e-value cut-off of 10^{-10} , a threshold of 30% amino acid sequence identity, and 60% coverage of both the query and the hit in the alignments.

For comparative genomics analysis, we retrieved the 12 *B. nordii* genomes along with their detailed information (Table S2), available at NCBI GenBank/RefSeq [22] or PATRIC [23] databases. To predict the AMR genes, Resistance Gene Identifier (RGI) module v 5.1.1 of the Comprehensive Antibiotic Resistance Database (CARD) database [24] and ResFinder v4.1 [25] were used with default parameters. PHASTER (PhAge Search Tool Enhanced Release) webserver [26] was used for the prediction of prophages. The genome was mined for insertion sequences (ISs) using ISEScan v1.7.2.3 [27] with an E-value cut-off of 10^{-10} . All the visualization and statistical analyses were performed in an R environment (<https://www.R-project.org>).

2.8. Pan-genome analysis

Pan- and core-genome analyses of thirteen genomes were performed using BPGA (Bacterial Pan Genome Analysis) v1.3 tool [28]. USEARCH [29] was used as a default clustering algorithm with a 50% sequence identity threshold for clustering orthologous proteins. A functional module implemented in BPGA v1.3 was used to determine the COG distribution of the pan-genome.

2.9. Selective pressure analysis of AMR genes

The AMR genes present in at least three genomes were used for the selective pressure analysis. Protein sequences of AMR genes were aligned using the Clustal Omega v1.2.4 [30] with default parameters. The codon alignment of nucleotide sequences was constructed by PAL2NAL [31] based on protein alignments. The codon alignment was used to calculate Ka and Ks using the Nei–Gojobori (NG) method [32] implemented in KaKs_Calculator v1.2 [33] with default parameters. The selective pressure (ω) was calculated as a ratio of non-synonymous (Ka) to synonymous (Ks) substitution rates.

2.10. Horizontal gene transfer analysis of AMR genes

To determine the AMR genes that were potentially acquired through HGT, a homology/taxonomy approach was employed using HGTector v2.0 with an E-value cut-off of 10^{-20} , a threshold of 30% amino acid sequence identity, 60% coverage of the query in the alignment, and

default bandwidth. The species *B. nordii* and the genus *Bacteroides* were set as self-group and close group, respectively.

3. Results

3.1. Phenotypic and genotypic antimicrobial resistance in MDR *B. nordii* PGMM4098

B. nordii PGMM4098 was phenotypically resistant to piperacillin-tazobactam, cefoxitin, metronidazole, and clindamycin. PCR assay confirmed the presence of *cfxA*, *nimE*, and *ermF* genes conferring resistance to beta-lactams (piperacillin-tazobactam and cefoxitin), metronidazole, and clindamycin, respectively.

3.2. General genome characteristics

The draft genome of *B. nordii* PGMM4098 has a size of almost 6 Mbp (5,714,214 bp) with a DNA G + C content of 40.80%. The general genomic characteristics of the MDR *B. nordii* PGMM4098 are summarized in Table 1 and Dataset S1.

3.3. Core and accessory genome variation

A total of thirteen genomes of *B. nordii* (including *B. nordii* PGMM4098) have been used for the pan-genome analysis for a better understanding of the evolution of this pathogenic bacterium. The pan-genome size of *B. nordii* contained 8495 gene clusters, whereas its core-genome size harbored 2649 gene clusters, which constitutes ~31% of the pan-genome (Fig. 1). The proportional fraction of the non-core and core-genome is 2.20; the high proportional value indicates the genome plasticity of *B. nordii*. This is in corroboration with the expansion of the *B. nordii* pan-genome, which currently is open but may close soon (Table S1). The pan-genome of *B. nordii* has further been subjected to COG functional analysis (Figs. 2 and S1). COG analysis revealed “Metabolism” as the most abundant COG category of the core genes, whereas the accessory and unique components are enriched with “Information storage and processing” COG function (Fig. S1). The “Poorly Characterized” COG category was excluded from further analysis. “Amino Acid Transport and Metabolism [E]” and “Carbohydrate Transport and Metabolism [G]” have been found as the most abundant COG classes in the core component of *B. nordii*. “Cell Wall/Membrane/Envelope Biogenesis [M]” was the most abundant COG class, followed by “Replication, Recombination, and Repair [L]” in the accessory and unique components of the pan-genome. Importantly, COG class “Defense Mechanisms [V]” which generally includes AMR genes, has been found in higher proportion in accessory and unique components of the pan-genome as compared to the core genome (Fig. 2).

3.4. Phylogenetic analysis

BLAST analysis of the 16S rRNA gene sequence of the strain

Table 1
General genome characteristics of *B. nordii* PGMM4098.

S. No.	Feature	<i>B. nordii</i> PGMM4098
1	Number of contigs	67
2	Genome size (bp)	5,714,214
3	G + C content (%)	40.8
4	Number of genes	4733
5	Proteins with predicted functions	2849
6	Hypothetical proteins	1795
7	rRNA genes	5
8	tRNA genes	60
9	Proteins with COG annotations	2616
10	Proteins with KEGG annotations	1784
11	Membrane transporters	396
12	Prophage regions	2

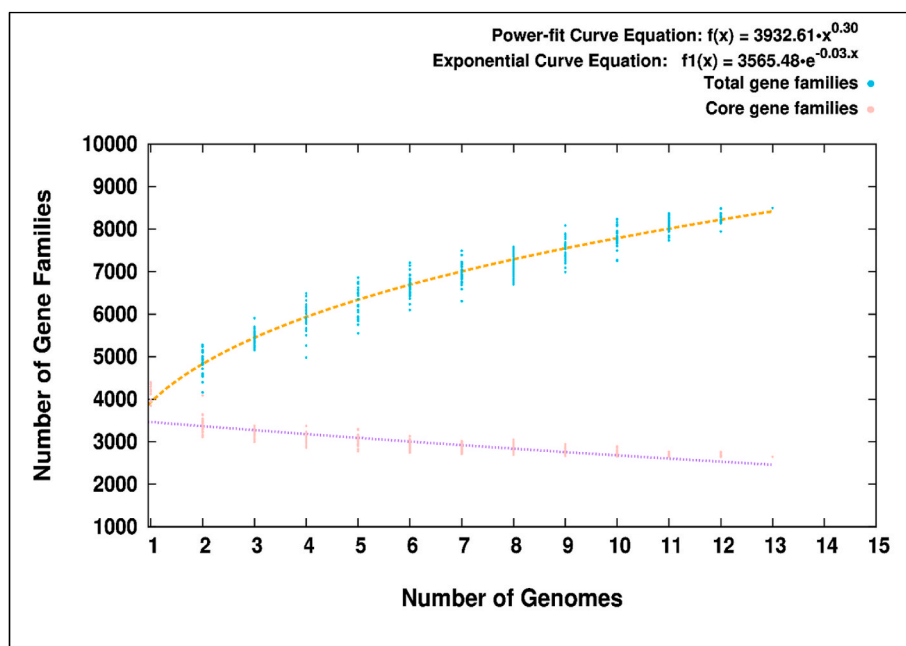


Fig. 1. A plot of the pan- and core-genomes of the species *Bacteroides nordii* (n = 13 members). The plot represents a stabilized core structure but an open pan-genome.

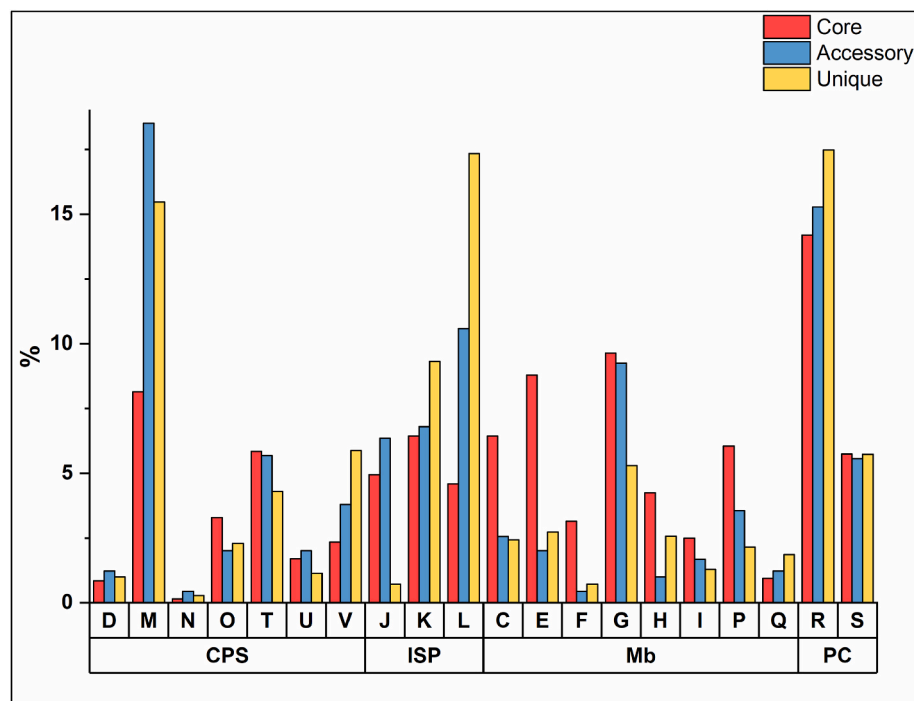


Fig. 2. The bar plots of the different COG classes present in the pan-genome of *Bacteroides nordii*.

CPS: Cellular Processes and Signaling; ISP: Information Storage and Processing; Mb: Metabolism; PC: Poorly Characterized; D: Cell cycle control, cell division, chromosome partitioning; M: Cell motility; N: Cell wall/membrane/envelope biogenesis; O: Post-translational modification, protein turnover, chaperones; T: Signal transduction mechanisms; U: Intracellular trafficking, secretion, and vesicular transport; V: Defense mechanisms; A: RNA processing and modification; J: Translation, ribosomal structure, and biogenesis; K: Transcription; L: Replication, recombination, and repair; C: Energy production and conversion; E: Amino acid transport and metabolism; F: Nucleotide transport and metabolism; G: Carbohydrate transport and metabolism; H: Coenzyme transport and metabolism; I: Lipid transport and metabolism; P: Inorganic ion transport and metabolism; Q: Secondary metabolites biosynthesis, transport, and catabolism; R: General function prediction only; S: Function unknown.

PGMM4098 members with the other members of the species *B. nordii*, demonstrated at least 99.67% sequence identity (Table S2). However, the 16S rRNA gene sequence of the strain PGMM4098 could not be compared with all twelve strains of *B. nordii* due to missing 16S rRNA genes in a few genomes. Thus, the phylogenetic positioning of strain PGMM4098 was determined by a robust phylogenetic method using the core-proteome (Fig. 3). The core proteome comprises 2649 proteins which are conserved in 13 strains of *B. nordii*. The phylogeny revealed clustering of *B. nordii* PGMM4098 with *B. nordii* ERR1600577-bin.6 strain. The *B. nordii* ERR1600577-bin.6 strain was recovered from

human gut metagenome associated with metabolic diseases such as Type 2 diabetes (T2D) and T2D-related comorbidities; however, the exact role of this strain remains unknown.

3.5. Antimicrobial resistance

To determine the genotypic AMR profile, thirteen *B. nordii* genomes were mined using the CARD and ResFinder databases. *In silico* AMR profile of thirteen *B. nordii* strains is shown in Fig. 4. In total, thirteen different AMR genes have been predicted in the genomes of *B. nordii*

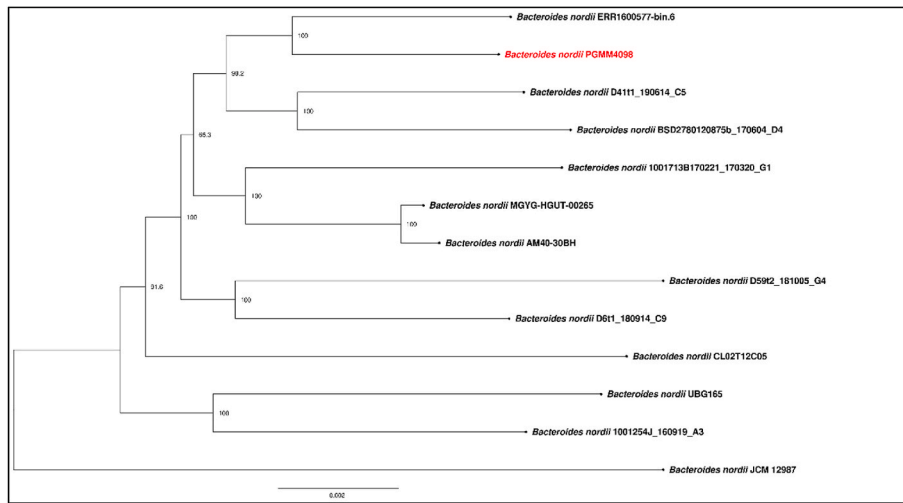


Fig. 3. A core-proteome-based phylogenetic tree of 13 members of the species *Bacteroides nordii*, including *Bacteroides nordii* PGMM4098. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) are shown next to the branches. Strain PGMM4098 (red font) is clustering with the strain ERR1600577-bin.6 of the species *Bacteroides nordii*. The phylogenetic tree is constructed by the maximum likelihood method. Bootstrap percentages above 50 are shown at node points, based on 1000 replicates of the phylogenetic tree. The scale bar corresponds to 2 substitutions per 1000 positions for a unit branch length.

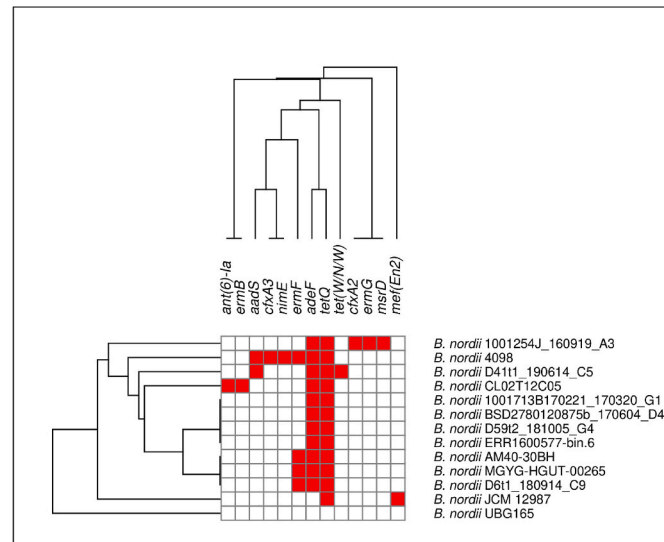


Fig. 4. Heatmap demonstrating the genotypic antimicrobial resistance profile of thirteen *Bacteroides nordii* strains. The red cell represents the presence of the gene, whereas a blank cell represents the absence of the gene.

strains. The *adeF* and *tetQ* are the only AMR genes that were present in at least 80% ($n = 11$) of the total genomes. Among all the strains, *B. nordii* PGMM4098 had the maximum number of AMR genes ($n = 6$), including *aadS* (streptomycin resistance), *adeF* (multidrug efflux pump), *cfxA3* (β -lactam resistance), *ermF* (macrolide-lincosamide-streptogramin B resistance), *nimE* (metronidazole resistance), and *tetQ* (tetracycline resistance) encoded in its genome. Importantly, *B. nordii* PGMM4098 was the only strain with the *nimE* gene in its genome. *B. nordii* UBG165 was the only strain devoid of any AMR gene.

3.6. Horizontal gene transfer of antimicrobial resistance genes

HGT was evaluated for all the AMR genes encoded in the genome of *B. nordii* strains. Only four genes, including *nimE*, *aadS*, *mef(En2)*, and *ermB/F/G*, conferring metronidazole, streptomycin, macrolide, and macrolide-lincosamide-streptogramin B resistance, respectively, have been found to be acquired via the process of HGT. The *aadS* and *mef(En2)* genes were found to be acquired from the phylum Bacteroidetes, whereas the order Bacteroidales was the donor taxon of *nimE* gene. The donor taxon of *ermB/F/G* gene remains unknown. Further, no insertion

sequence or phage element carrying these AMR genes was detected in the genome of *B. nordii* PGMM4098.

3.7. Selective pressure operating on antimicrobial resistance genes

The three AMR genes viz., *ermB/F/G*, *tetQ*, and *cfxA2/A3* conferring macrolide-lincosamide-streptogramin B, tetracycline, and β -lactam resistance, respectively, have been found under the operation of positive selective pressure with ω values of 1.52, 1.10, and 1.41, respectively. The $\omega > 1$ indicates a higher non-synonymous to synonymous mutation rate and suggests a prevalence of diversifying selection of these genes. In contrast, *adeF* gene encoding the antibiotic efflux pump was found to be under negative selective pressure ($\omega = 0.76$). The $\omega < 1$ indicates stringent purifying or stabilizing selection acting against amino acid changes in the encoding gene.

4. Discussion

Our understanding of the genome biology and evolution of *B. nordii* remains poor, given the low number of available genomes of this pathogenic species. To this end, we, for the first time, reported and characterized the genome of an MDR strain of *B. nordii* from a clinical sample. To explore the evolution of this species, we determined the pan-genome of *B. nordii*. We demonstrated its open nature, which is attributed to the higher proportion of accessory and unique genes than core genes. The open pan-genome indicates a continuous expansion of *B. nordii*, which can be attributed to the accumulation of specific functions in the genome via different gene acquisition strategies. The gain and loss of certain functions is the primary driving force behind the species expansion, which imparts microbes an adaptive ability to a wide range of ecological environments [10].

Given the human gut as a major habitat of *B. nordii*, metabolism-related functions are profusely required for adaptive purposes in such a carbohydrate and protein-rich environment. Our analysis also found carbohydrate and amino acid metabolism as the most abundant functions encoded by the core and non-core genes (accessory and unique genes) under the “Metabolism” COG category. This is consistent with the previous studies [10,34–36] which demonstrated that the gut anaerobes are primarily involved in the degradation of complex polymers viz., carbohydrates and proteins into smaller units, which are further metabolized for deriving energy. Notably, the non-core genes are of utmost importance owing to their role in adaptive functions. Acquisition of non-core genes occurs mainly via HGT [10] for environmental adaptation and thus suggests the adaptive potential of *B. nordii*. This is also in line with previous findings that the metabolic genes undergo HGT

more frequently than other informational genes [37].

The open nature of the pan-genome is also attributed to the complexity of a given environment in which the bacteria inhabit [38]. Microbes from complex and varied environments are known to possess an open pan-genome [10,12]. Complex environments such as human gut allow HGT of certain traits among different bacterial communities leading to bacterial genome evolution in terms of antibiotic resistance, virulence emergence, and compatibility to environmental stresses [39]. Although there is no well-documented report on infections caused by *B. nordii*, a few gut metagenomic studies have demonstrated the role of *B. nordii* in mild and severe diseases suggesting its pathogenic potential [40]. In a recent study, *B. nordii* was found to be enriched in the gut microbiome of antibiotic-naïve patients with Coronavirus Disease 2019 [41]. Similarly, *B. nordii* was found in high abundance in the gut microbiome of ankylosing spondylitis patients [42]. The gut microbiome is a complex environment that promotes the growth of sympatric microbes leading to the open pan-genome, as demonstrated in a recent study on gut Bifidobacteria [10]. Thus, niche-complexity can be reasoned for the open pan-genome of *B. nordii*, which suggests a continuous evolution of this species in a given environment. However, a limited number of genomes in our study necessitates the re-analysis of the pan-genome of *B. nordii* with additional genomes.

Generally, antimicrobial resistance and defense mechanisms are not the core or housekeeping functions of bacteria and constitute the accessory component of the pan-genome, which is acquired via the process of HGT [39,43]. This is consistent with the functional analysis of the pan-genome of *B. nordii*, which revealed a higher proportion of COG class “Defense Mechanisms [V]” in the accessory and unique components as compared to the core-genome. This is consistent with a previous study which demonstrated that most of the antimicrobial determinants of *E. coli* belong to the accessory component and are classified under the COG class “Defense Mechanisms [V]” [44]. Similar to this, thirteen AMR genes predicted in our analysis constitute the non-core component of the pan-genome as none of these genes is present in all thirteen strains of *B. nordii*. The AMR distribution in *B. nordii* with one strain harboring no AMR determinant while another possessing six AMR genes suggests the genomic diversity of this species. Thus, the presence of AMR genes as an accessory component suggests the adaptive functions of *B. nordii* in a human host environment burdened with a high antimicrobial load. The three AMR genes, including *nimE*, *aadS*, and *mef(En2)*, conferring metronidazole, streptomycin, and macrolide resistance were found to be acquired via the process of HGT from anaerobic Bacteroidetes, whereas the donor taxon of *ermB/F/G* gene conferring macrolide-lincosamide-streptogramin B resistance remains unknown. Besides, the mode of HGT remains unknown given that no insertion sequence or phage element carrying these AMR genes was detected in the genomes of *B. nordii*. Intra-anaerobe HGT suggests the adaptation of *B. nordii* to the anaerobic and stressful environment of human gut. This is in corroboration with a recent study which demonstrated elevated rates of HGT in the human gut microbiome [45].

AMR has emerged as one of the major public health problems of the 21st century. Antimicrobial-resistant organisms are ubiquitous to humans, animals, and the environment. Unsolicited usage of antimicrobials exerts continuous selective pressure on the AMR genes leading to the selection of novel mutations and resistant phenotypes. Further, AMR transmission from the environment or animal to the human host makes this condition more critical. For instance, the tetracycline resistance gene was found to be under the positive selective pressure in ruminal bacteria [46]. Similarly, another study found genes (*bla_{IMP}*, and *bla_{VIM}*) encoding metallo-β-lactamases under the operation of positive selective pressure [47]. To this end, we demonstrated that *ermB/F/G*, *tetQ*, and *cfxA2/A3* genes of *B. nordii* are under the positive selective pressure indicating the diversification of these AMR genes. This might cause the evolution of erythromycin, tetracycline, and β-lactam resistance in *B. nordii* leading to the emergence of new resistant strains in the near future. In contrast, negative selective pressure operating on *adeF*

gene of *B. nordii* indicates purifying or stabilizing selection acting against amino acid changes in the encoding gene. The selective pressure could not be calculated for the remaining AMR genes since at least three orthologous copies of a given AMR gene are required for this analysis.

5. Conclusion

The study presents the comprehensive genome characterization of *B. nordii* PGMM4098, the first MDR strain of *B. nordii* isolated from a clinical sample. Notably, *B. nordii* PGMM4098 is the only strain of this species harboring the *nimE* gene, which confers metronidazole resistance. AMR distribution in *B. nordii* is reflected in the evolutionary analysis, which has demonstrated an open pan-genome of this species, indicating the ability to acquire new genes leading to the emergence of new phenotypes and strains. To this end, three AMR genes, including *nimE*, *aadS*, and *mef(En2)*, conferring metronidazole, streptomycin, and macrolide resistance, respectively, have been predicted to be acquired by *B. nordii* through the process of intra-anaerobic HGT suggesting the donor specificity and environmental adaptation of *B. nordii* in the human gut. Further, *ermB/F/G*, *tetQ*, and *cfxA2/A3* conferring macrolide-lincosamide-streptogramin B, tetracycline, and β-lactam resistance, respectively, were found under the diversifying selective pressure which might lead to the emergence of new resistant strains of *B. nordii* in the near future. Thus, the judicious use of antimicrobials is the need of the hour, and empirical management needs to be slowed down. Overall, the study highlights the application of WGS for studying the prevalence and genetic basis of AMR evolution in understudied anaerobic pathogens.

Data availability

This whole-genome shotgun project has been deposited at NCBI GenBank under the accession JAJNGY000000000. Besides, the whole genome data is also available at the Figshare repository (<https://doi.org/10.6084/m9.figshare.19936025.v1>).

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CRediT authorship contribution statement

Vikas Sharma: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Anshul Sood:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Pallab Ray:** Writing – review & editing, Funding acquisition, Conceptualization. **Archana Angrup:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Archana Angrup reports financial support was provided by Indian Council of Medical Research.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mpa.2022.105811>.

[org/10.1016/j.micpath.2022.105811](https://doi.org/10.1016/j.micpath.2022.105811).

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